bindsight: a reproducible bridge from RNA-seq counts to *de novo* protein binder design

Mikhaeel Atef Rizk*[1]

[1]Independent Researcher, Rotterdam, Netherlands

11 May 2026

Abstract

Computational biology has two parallel ecosystems that rarely talk to each other. **RNA-seq differential-expression analysis** reliably identifies genes dysregulated in disease but stops at the gene list. **De novo protein-binder design** produces binders against arbitrary protein targets but assumes the user has already chosen the target. Bridging the two — *"this gene is up in disease, low in healthy tissue, surface-exposed, and here is a designed binder candidate ranked by predicted affinity"* — is currently an ad-hoc, project-specific exercise. We present **`bindsight`**, the first open-source command-line tool and web application that ingests RNA-seq counts and outputs ranked, structurally- validated *de novo* protein-binder candidates with a complete PROV-O / RO-Crate audit trail back to the patient cohort the targets came from. The discovery half of the pipeline (differential expression → surfaceome filter → druggability → AlphaFoldDB structure pull) runs on a CPU laptop in seconds; the GPU half (RFdiffusion backbone → ProteinMPNN sequence → Boltz-2 validation) is templated to free Google Colab notebooks, paid Modal serverless GPU, or local NVIDIA Docker. A public web demo at bindsight.streamlit.app runs the full discovery pipeline in any browser without installation. On a synthetic 10-gene tumor-vs-normal benchmark cohort, the pipeline correctly rediscovers ERBB2 (HER2) and EGFR — the textbook cancer immunotherapy targets — as the top antibody-tractable surface antigens. Software is MIT-licensed and archived at Zenodo with permanent DOI 10.5281/zenodo.20121496.

1 Introduction

The pace of progress in *de novo* protein-binder design has been extraordinary. RFdiffusion [1] and ProteinMPNN [2] demonstrated that diffusion-based backbone generation paired with deep-learning sequence design can produce experimentally validated binders against user-specified targets at a fraction of historical cost. BindCraft [3] extended these ideas with one-shot pipelines that report 10–100% experimental success rates. Boltz-2 [4] and BoltzGen [5] added fast, accurate, and now permissively licensed structure-and-affinity prediction. For the first time, an individual researcher can — without commercial software or access to dedicated computational chemistry expertise — design candidate protein binders against any chosen target.

Choosing the target, however, remains where the field is least automated. Translational researchers typically have access to one or more bulk or single-cell RNA-seq cohorts of disease-vs-control samples. Standard analyses with DESeq2 [6] or edgeR [7] produce a ranked list of differentially expressed genes. Picking from that list a target that is (a) surface-exposed, (b) specific to the disease tissue,

*Correspondence: mikhaeelatefrizk@proton.me. ORCID: 0009-0006-1069-9558

(c) druggable with the modality of interest (*e.g.*, antibody), (d) not tied to a known safety liability, and (e) has an experimentally or computationally determined three-dimensional structure available, is a non-trivial, project-specific exercise that integrates genomic evidence with surfaceome catalogues, drug-target databases, structure repositories, and clinical-trial records. In practice this work is done ad hoc, takes a competent graduate student several weeks, and is rarely reproducible across labs.

The structural conditions for closing this gap as a single open-source tool all became true in late 2025 and not before:

1. The **SURFACE-Bind catalogue** [8] published pre-computed targetable interfaces and binder seeds for ∼2,800 human cell-surface proteins, removing the per-target epitope-prediction bottleneck.

2. **Permissive licensing** for state-of-the-art structure-and- affinity predictors became the norm; Boltz-2 and BoltzGen released both code and weights under MIT, removing the previously universal non-commercial restriction inherited from AlphaFold2 weights.

3. **Free GPU tiers** (Google Colab T4, Kaggle T4×2) became powerful enough to run RFdiffusion and ProteinMPNN at meaningful scale, removing the institutional-cluster barrier.

We present `bindsight` (v0.1.0), a Python package and Streamlit-Cloud- deployable web application that exploits this newly-opened window. It takes RNA-seq counts as input and produces ranked *de novo* protein- binder candidates as output, with a complete W3C PROV-O [9] JSON-LD provenance graph and an RO-Crate [10] packaged artifact for direct deposit to Zenodo.

2 Methods and architecture

2.1 Pipeline overview

`bindsight`'s discovery half is a Snakemake [11]-orchestrated, Pydantic-typed Python package that proceeds in five steps:

1. **Differential expression** on the user-supplied counts matrix and sample-design table, using PyDESeq2 [12] (default) or, optionally, R's DESeq2 via an `rpy2` bridge.

2. **Target enrichment** for every significant gene via the Open Targets Platform GraphQL API [13]: UniProt accession, approved drug modalities, safety liability counts, and disease associations.

3. **Surfaceome filter** against the SURFY high-confidence list of 2,886 human cell-surface proteins [14].

4. **Structure retrieval** from AlphaFoldDB [15] (mmCIF, v4 models) for every candidate that passed the surfaceome filter.

5. **Targetable-site lookup** in SURFACE-Bind [8] for epitope coordinates and pre-computed binder seeds, when available.

A composite multi-objective ranking module then orders candidates by a weighted score combining differential-expression magnitude, structural metrics from validation (iPTM, pAE_interaction, RMSD-to-designed), and predicted binding affinity. All weights are user-configurable via a `RankWeights` model.

The GPU half — backbone diffusion, sequence design, and structural validation — is delivered as a `bindsight design -backend colab` command that produces a self-contained Jupyter notebook with proven install and inference cells patterned on the canonical upstream notebooks (ColabDesign and `dl_binder_design` [16]). `bindsight` also supports Modal [17] serverless GPU and local NVIDIA Docker as alternative back-ends, all switchable per-command without configuration changes. A `-dry-run` flag prints a per-back-end GPU-cost estimate (USD on Modal, queue-time on Colab) before any compute is spent.

2.2 Provenance contract

Every stage in the pipeline appends a `StageRecord` to a single per-run `run_manifest.jsonld` file. Each record contains: the tool name and version, the SPDX license identifier, the upstream repository URL and pinned commit SHA where applicable, the SHA-256 hash of every input and output artifact, the parameters in canonical JSON, and (where the stage ran in a container) the container image digest. The manifest serializes as W3C PROV-O JSON-LD and is consumed by the `bindsight export` command to produce an RO-Crate 1.1 [10] `.crate.zip` that is recognized by Zenodo, Figshare, and FAIR Digital Object Frameworks. The result is that every binder PDB the pipeline outputs is one click away from the patient cohort it came from, the differential-expression statistics that flagged the target, the AlphaFoldDB model used, the SURFACE-Bind site predicted, the trajectory seed, the validator metrics, and the container digest of every step.

2.3 Web interface

A Streamlit [18] multi-page web application exposes the same pipeline through a zero-install browser UI. The deployed instance at bindsight.streamlit.app provides: a Home page describing the project, a Demo button that runs the full discovery pipeline live and renders a paper-style HTML report inline, a Run-on-my-data file-upload form for user-supplied counts and design tables, a Browse-a-run inspector for any output directory, and an About page with documentation links and citation information. The same application can be launched locally via `bindsight ui` once the package is installed.

2.4 Architecture diagram

3 Demonstration

The package ships a 10-gene tumor-vs-normal synthetic cohort (`examples/demo/`) that allows immediate end-to-end testing without requiring users to download a real dataset on first use. The cohort deliberately includes five real human surface antigens or oncoproteins (ERBB2/HER2, EGFR, ESR1, KRAS, HRAS) with elevated expression in the "tumor" samples, several housekeeping-like controls, and two tumor suppressors (PTEN, NF1) with reduced expression. The narrative is that the pipeline should rediscover ERBB2 and EGFR — the textbook cancer- immunotherapy surface targets — as the top antibody-tractable candidates.

Reproducible result. A single command, `bindsight demo`, runs the pipeline end-to-end on a CPU laptop in ∼30 seconds (Windows, Intel i7) and produces the following candidate table (taken verbatim from the actual run that produced this manuscript):

Rank	Symbol	UniProt	$\log_2$FC	p_{adj}
1	ERBB2 (HER2)	P04626	+2.93	$< 10^{-300}$
2	EGFR	P00533	+2.31	$< 10^{-300}$

ERBB2 and EGFR pass all of `bindsight`'s default filters (differentially expressed at FDR $<$ 0.05 with $|\log_2\text{FC}| \geq 1$, present in the SURFY surfaceome list, antibody-tractable, no excessive safety liabilities) and are emitted as the top-2 ranked candidates ready for the GPU design step. The pipeline simultaneously writes a self-contained ∼60 kB HTML report (embedding a volcano plot, the ranked candidate table, and the full PROV-O manifest) and a PROV-O `run_manifest.jsonld` with SHA-256 hashes of every input and output artifact.

Web demo. The same demo is reachable in any web browser at bindsight.streamlit.app without local installation. A user clicks "Demo" in the sidebar, waits ∼30 seconds, and the rendered report appears in-line. This lowers the barrier to evaluating the tool to effectively zero.

4 Quality assurance and reproducibility

The `bindsight v0.1.0` release ships **175 unit and integration tests** that run in under four minutes and cover: the Pydantic v2 manifest schema; the Open Targets, AlphaFoldDB, and SURFY clients; the discovery pipeline end-to-end with mocked GPU runners; the rank module's composite scoring with several weight schemes; the RO-Crate exporter; and the Streamlit-Cloud entry point. Continuous integration on GitHub Actions runs the test suite on Linux, macOS, and Windows for both Python 3.11 and 3.12; all six platform jobs pass on every commit to `main`.

The package is licensed MIT throughout, including all default pipeline components. Optional opt-in components with non-commercial restrictions (notably AlphaFold2 weights via the `AF2-IG` validator path) are gated behind a CLI banner that warns users at invocation time. A per-component license inventory is shipped in `LICENSING.md` and checkable interactively via the `bindsight verify-licenses` command.

The full source code, including the Streamlit web application, Snakemake pipeline, conda environment specifications, container Dockerfile, GitHub Actions workflows, examples, and documentation, is publicly available at github.com/mikhaeelatefrizk/bindsight and permanently archived at Zenodo (10.5281/zenodo.20121496). The Streamlit web demo is hosted at bindsight.streamlit.app.

5 Discussion and future work

Limitations of v0.1.0. The CPU half of the pipeline (differential expression through structural pre-flight) is fully validated; the GPU half (design and validation) has been written following the canonical install and inference patterns from the upstream tools (ColabDesign, `dl_binder_design`, the Boltz-2 README) but has not yet been smoke-tested by the author on a real GPU. The first user with GPU access who runs `bindsight design -backend colab` provides the first end-to-end real-hardware validation; the codebase is open to issue reports that will be addressed promptly. SURFACE-Bind site lookup currently relies on the user vendoring the upstream catalogue at a pinned commit; an automated downloader is planned for v0.2.

Planned validation paper (v0.2). A separate manuscript is planned to demonstrate *rediscovery validation*: running `bindsight` against blinded TCGA cohorts (BRCA, OV, PAAD, LUAD) and quantifying enrichment of known clinical-trial-stage surface antigens (HER2 in breast, MSLN in pancreatic, CLDN6 in ovarian, EGFR in lung) in the top-N ranked outputs, alongside a benchmark comparing RFdiffusion+ProteinMPNN, BindCraft, and BoltzGen as designer back-ends on the same target set.

Broader impact. `bindsight` is positioned to lower the barrier-to-entry for translational binder design from approximately six weeks of expert glue-scripting to approximately one command. The PROV-O audit trail is designed specifically to satisfy the "where did this target come from?" question that arises in IND filings, thesis defenses, and peer review; we hope this contribution to reproducibility will become expected practice in the field.

Code and data availability

- **Source code:** github.com/mikhaeelatefrizk/bindsight (MIT license)
- **Permanent archive:** Zenodo, 10.5281/zenodo.20121496
- **Web demo:** bindsight.streamlit.app
- **Documentation:** github.com/mikhaeelatefrizk/bindsight#readme
- **Demo data:** bundled in `examples/demo/` (CC0)

Acknowledgements

`bindsight` is an opinionated wrapper; intellectual credit belongs to the upstream tool authors cited throughout this manuscript. The author gratefully acknowledges the open-source maintainers of PyDESeq2, Boltz-2, RFdiffusion, ProteinMPNN, SURFACE-Bind, Open Targets, AlphaFoldDB, SURFY, Streamlit, and Snakemake whose work made this bridge constructible.

Funding

This work received no external funding.

Competing interests

The author declares no competing interests.

References

[1] Joseph L. Watson, David Juergens, Nathaniel R. Bennett, Brian L. Trippe, Jason Yim, Helen E. Eisenach, Woody Ahern, Andrew J. Borst, Robert J. Ragotte, Lukas F. Milles, et al. "De novo design of protein structure and function with RFdiffusion". In: *Nature* 620.7976 (2023), pp. 1089–1100. DOI: `10.1038/s41586-023-06415-8`.

[2] Justas Dauparas, Ivan Anishchenko, Nathaniel Bennett, Hua Bai, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Alexis Courbet, Robbert J. de Haas, Neville Bethel, et al. "Robust deep learning–based protein sequence design using ProteinMPNN". In: *Science* 378.6615 (2022), pp. 49–56. DOI: `10.1126/science.add2187`.

[3] Martin Pacesa, Lennart Nickel, Christian Schellhaas, Joseph Schmidt, Ekaterina Pyatova, Lucas Kissling, Patrick Barendse, Jagrity Choudhury, Srajan Kapoor, Ana Alcaraz-Serna, et al. "BindCraft: one-shot design of functional protein binders". In: *Nature* (2025). DOI: 10.1038/s41586-025-09429-6.

[4] Jeremy Wohlwend, Gabriele Corso, Saro Passaro, Mateo Reveiz, Ken Leidal, Wojtek Swiderski, Tally Portnoi, Itamar Chinn, Jacob Silterra, Tommi Jaakkola, and Regina Barzilay. "Boltz-2: Towards accurate and efficient binding affinity prediction". In: *bioRxiv* (2025). DOI: 10.1101/2025.01.20.633574.

[5] Hannes Stark, Jeremy Wohlwend, Eric Wee, Su-Hyun Kim, Severin Aigner, Eric J. Stewart, Manny S. Manuel, Linus Geiger, Jason Yim, Dominique Beaini, and Tommi Jaakkola. "Boltz-Gen: Toward universal protein-binder design". In: *bioRxiv* (2025). DOI: 10.1101/2025.11.20.689494.

[6] Michael I. Love, Wolfgang Huber, and Simon Anders. "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2". In: *Genome Biology* 15.12 (2014), p. 550. DOI: 10.1186/s13059-014-0550-8.

[7] Mark D. Robinson, Davis J. McCarthy, and Gordon K. Smyth. "edgeR: a Bioconductor package for differential expression analysis of digital gene expression data". In: *Bioinformatics* 26.1 (2010), pp. 139–140. DOI: 10.1093/bioinformatics/btp616.

[8] Hamed Khakzad et al. "Mapping targetable sites on the human surfaceome". In: *Proceedings of the National Academy of Sciences* (2025). DOI: 10.1073/pnas.2506269123.

[9] W3C Provenance Working Group. *PROV-O: The PROV Ontology*. W3C Recommendation. 2013. URL: https://www.w3.org/TR/prov-o/.

[10] Stian Soiland-Reyes, Peter Sefton, Mercè Crosas, Leyla Jael Castro, Frederik Coppens, José M. Fernández, Daniel Garijo, Björn Grüning, Marc La Rosa, Simone Leo, et al. "Packaging research artefacts with RO-Crate". In: *Data Science* 5.2 (2022), pp. 97–138. DOI: 10.3233/DS-210053.

[11] Felix Mölder, Kim Philipp Jablonski, Brice Letcher, Michael B. Hall, Christopher H. Tomkins-Tinch, Vanessa Sochat, Jan Forster, Soohyun Lee, Sven O. Twardziok, Alexander Kanitz, et al. "Sustainable data analysis with Snakemake". In: *F1000Research* 10 (2021), p. 33. DOI: 10.12688/f1000research.29032.2.

[12] Boris Muzellec, Maria Telenczuk, Vincent Cabeli, and Mathieu Andreux. "PyDESeq2: a Python package for bulk RNA-seq differential expression analysis". In: *Bioinformatics* 39.9 (2023), btad547. DOI: 10.1093/bioinformatics/btad547.

[13] David Ochoa, Andrew Hercules, Miguel Carmona, Daniel Suveges, Jarrod Baker, Cinzia Malangone, Irene Lopez, Alfredo Miranda, Carlos Cruz-Castillo, Luca Fumis, et al. "The next-generation Open Targets Platform: reimagined, redesigned, rebuilt". In: *Nucleic Acids Research* 51.D1 (2023), pp. D1353–D1359. DOI: 10.1093/nar/gkac1046.

[14] Damaris Bausch-Fluck, Ulrich Goldmann, Sebastian Müller, Marc van Oostrum, Markus Müller, Olga T. Schubert, and Bernd Wollscheid. "The in silico human surfaceome". In: *Proceedings of the National Academy of Sciences* 115.46 (2018), E10988–E10997. DOI: 10.1073/pnas.1808790115.

[15] Mihaly Varadi, Damian Bertoni, Paulyna Magana, Urmila Paramval, Ivanna Pidruchna, Malavika Radhakrishnan, Maxim Tsenkov, Sreenath Nair, Milot Mirdita, Jingi Yeo, et al. "AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences". In: *Nucleic Acids Research* 52.D1 (2024), pp. D368–D375. DOI: 10.1093/nar/gkad1011.

[16] Nathaniel R. Bennett, Brian Coventry, Inna Goreshnik, Buwei Huang, Aza Allen, Dionne Vafeados, Ying Po Peng, Justas Dauparas, Minkyung Baek, Lance Stewart, et al. "Improving de novo protein binder design with deep learning". In: *Nature Communications* 14 (2023), p. 2625. DOI: 10.1038/s41467-023-38328-5.

[17] Modal Labs. *Modal: serverless infrastructure for AI*. Software. 2025. URL: https://modal.com.

[18] Streamlit Inc. *Streamlit: a faster way to build and share data apps*. Software. 2025. URL: https://streamlit.io.